

Interpretable Card-Play Strategies from Evolved Logic Networks

A Neurosymbolic Agent for Sueca

TODO Author One TODO Author Two TODO Author Three
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TODO University — TODO Course

Most strong card-playing AIs are black boxes — we asked whether competitive play can be human-auditable

The status quo

Deep PIMC, neural policy nets
Millions of floating-point weights
→ opaque
“Why did you play that card?”
“...the matrix says so.”

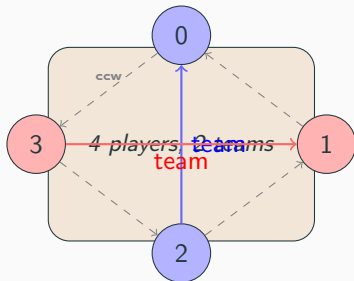
Our goal

Evolved logic gates
A handful of
human-readable
IF/THEN rules
“...the trick is cheap”
and I want to build a void.”

Thesis: Discrete logic gates + neuroevolution
→ competitive play *and* interpretability.

Sueca's distinctive rules create a rich strategic texture

Table layout

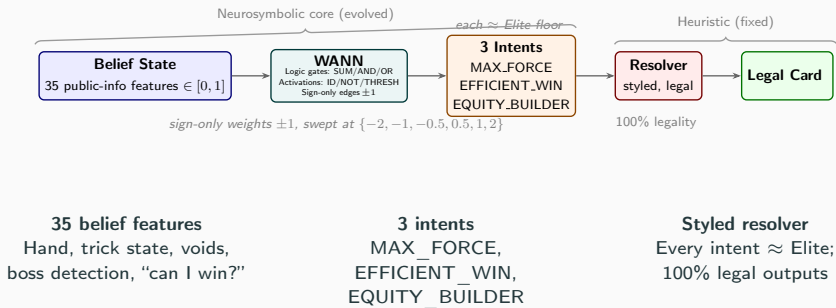


Key rules

- **40 cards** (no 8,9,10)
- Rank: **A > 7 > K > J > Q > 6-2**
- 7 = *manilha* (2nd-highest!)
- **120 points/deal**; game-point tiers
- Must **follow suit**; trump cuts
- **Public void tracking**
- Imperfect information:
you see only your hand + history

These rules define the strategic vocabulary the WANN must learn.

The agent reads a public-information belief state through evolved logic gates and outputs a strategic intent

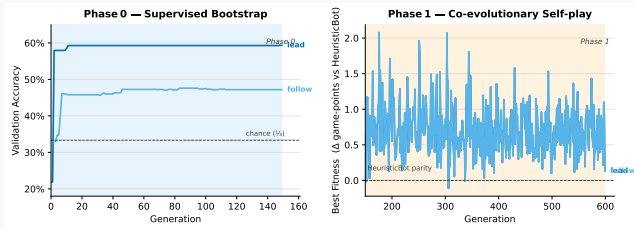


Weight-agnostic networks replace learned weights with evolved logic-gate topology

Node types		Why it compiles to rules
Aggregation	Activation	• Connections: sign only (± 1)
SUM ($\sum x_i$)	IDENTITY (x)	• Shared weight sweep: $W \in \{-2, -1, -0.5, 0.5, 1, 2\}$
MIN ($\min x_i$) = AND	NOT ($1-x$)	
MAX ($\max x_i$) = OR	THRESHOLD ($x > 0.5$)	
• No MEAN (precision), no SIGMOID (rules) • All outputs clamped to $[0, 1]$		• Topology must work at <i>all</i> scales • Every gate is a logic primitive → Boolean-ish rules • Tie-breaking: random among max intents

The architecture constraint is the interpretability guarantee.

Training bootstraps on expert demonstrations then switches to co-evolutionary self-play



Phase 0 (gens 0–149)

Supervised pretraining on PIMC data
PFS-NEAT from **zero** connections
Split: *lead* vs. *follow* datasets
Adaptive 2-stage mutation filter

Phase 1 (gens 150–599)

Co-evolutionary self-play
Dynamic per-decision brain routing
Delta-fitness with CRN
HOF + MAP-Elites opponent pools

A cheap supra-Elite teacher made the champion 4.5× simpler at equal strength

Problem diagnosed

- Deep PIMC **only tied** Elite
- Root cause: **myopic leaf eval** (raw score, no positional estimate)
- Depth doesn't fix a blind leaf

What the better teacher bought

	v5	v6 (canonical)
vs. Elite	52.7%	52.1%
Hidden gates	132	29
Connections	188	49
Teacher	alpha-beta	rollout

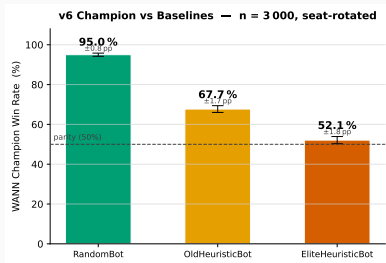
Stronger teacher → simpler champion.

The ceiling is the resolver, not the teacher.

Solution

- Flat Monte-Carlo PIMC + **Elite** **playouts**
- Rollout policy improvement: 1-ply + Elite $\geq$ Elite
- **62% vs. Elite**, $\sim 1000\times$ faster

The champion beats EliteHeuristicBot 52.1% $\pm$ 1.8% ($n = 3000$)



Tournament matrix (win%, $n = 3000$)

	Rnd	Old	Elite	WANN
Random	50.0	10.8	4.7	5.1
Old	89.2	50.0	32.5	32.3
Elite	95.3	67.5	50.0	47.9
WANN	95.0	67.7	52.1	50.0

- 95% CI [50.3%, 53.9%]
excludes 50%
- Card pts: 60.2 vs. 59.8
- WANN vs. Old: 67.7%
(Elite vs. Old: 67.5%)
- Prior best: **30.2%**

Three root-cause fixes turned a collapsed network into an Elite-beating champion



Problem

Network collapsed to
"always EFFICIENT_WIN"
Intents unequally weak
20% / 37% / 25%

Fix #1

Styled resolver
Every intent $\approx$ Elite
48% / 47% / 46%
Collapse $\rightarrow$ merely ties

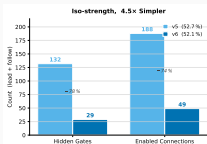
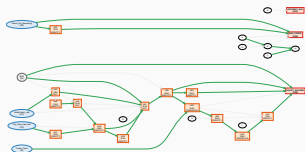
Fix #2 & #3

Best-of-3-intent labeling
Rollout teacher
62% vs. Elite
 $\rightarrow$ 4.5 $\times$ simpler

Diagnosed with a training-free harness (seconds), not blind retraining (hours).

The follow brain compiles to 5 logic gates at depth 5 — verified behavior-preserving

Network topology



Compiled rules (after folding)

```
hidden_42 = NOT(Holds_Boss_Led)
hidden_48 = (1 + Any_Opp_Void_Led)
hidden_41 = NOT((1 + h_42 + 2*hidden_48))
hidden_40 = THRESHOLD((h_41 + Trump_Count) > 0.5)
hidden_55 = THRESHOLD(hidden_40 > 0.5)
MAX_FORCE      = 2*Game_Pts_Remaining
EFFICIENT_WIN  = 0
EQUITY_BUILDER = 1 + 2*hidden_41 + hidden_55
```

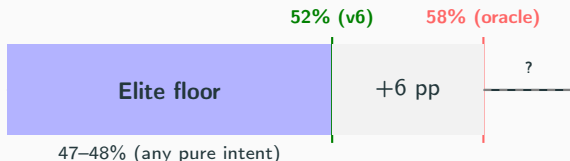
LEAD brain:

```
EQUITY = THRESHOLD(NOT(Trump_Count))
MAX_FORCE = (ahead & point-dense & partner-winning)
OR (no very-short side suit)
```

Verified on 2000 random states: removed symbols carry zero information. EFFICIENT_WIN = 0 in both brains

— the net never delegates to pure Elite.

The styled resolver that enabled the win also sets the strength ceiling — and we measured it



What limits us

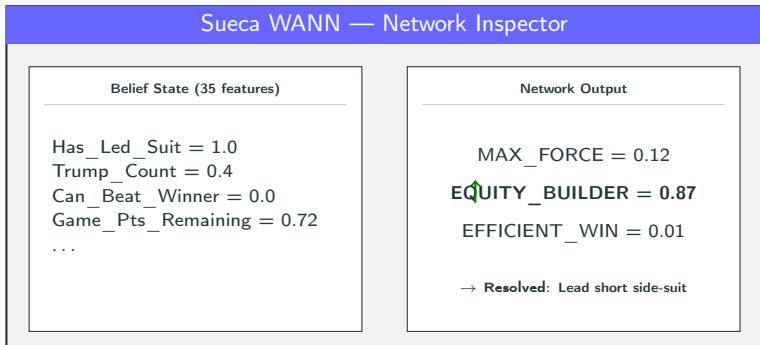
- 3 intents nearly equivalent in fitness
- Weak gradient for learned selection
- Lead decision hard: RF +2.5
- Follow learnable: RF +7.9

What could raise the cap

- More-differentiated intents
- Opponent-card inference features
- Knowledge distillation
- Richer belief state

Honest science: we measure our ceiling and report it. The +2% is the learned

You can explore the champion's decision-making interactively in the browser



React + WASM: Rust game engine compiled to WebAssembly
Play Sueca against the WANN — watch the logic flow in real time.

Richer intents and opponent-card inference are the path to breaking the 58% ceiling

Future directions

1. Richer intents

More-differentiated play styles
→ raise the oracle cap above 58%

2. Opponent-card inference

Can a logic circuit learn to infer hidden cards from optimal play?

3. Knowledge distillation

Champion → even smaller student
→ Pareto curve of strength vs. size

What we showed

- Evolved an **interpretable**, Elite-beating Sueca agent
- Verified pipeline: logic gates → **5 readable rules**
- Honest analysis of the resolver-floor ceiling
- Supra-Elite rollout teacher ($\sim 1000\times$ cheaper)
- All code, data, and rules are reproducible

Competitive, interpretable card-play AI is possible.

The rules fit on one slide.

Thank you. Questions?

Key references

Gaier & Ha 2019	Weight-Agnostic Neural Networks
Stanley & Miikkulainen 2002	NEAT
Mouret & Clune 2015	MAP-Elites
Reisinger et al. 2004	Coevolving modular (L-NEAT)

Reproducibility

All commands in README.md

Canonical champion: checkpoints/production/2026-06-14-2/
`cargo build -p sueca_wann -release && cargo test -all`

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